

Hermetic-Sheath Toroidal Pickup: BEM Calculation Log

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Abstract

Running log of the unit-current boundary-element calculation for the hermetic Meissner-sheath toroidal axion pickup, following the v2 task specification in `cc_task_spec.md`. We extract the dimensionless coefficients $F(\beta; \alpha) \equiv V_{\text{eff}}/R^3$ and $\ell(\beta; \alpha) \equiv L_p/(\mu_0 R)$ in the thin-annulus limit, test whether the favorable $Q \propto R_t^4$ scaling at fixed magnet cost survives parasitic inductance from slit and external return, and compute the crossover radius R_t^* at which the toroid figure of merit equals the long-solenoid Core baseline.

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1 Setup and goal

We follow the notation of *Hermetic-Sheath Toroidal Axion Pickup* (Safdi 2026, included as `toroid_optimization.pdf` in this repository). All lengths are in units of the major radius R_t ; $\mu_0 \equiv 1$; unit current $I \equiv 1$.

The magnet bore is

$$V_{\text{mag}} = \{1 < s < 1 + \beta, 0 < z < \alpha, 0 < \phi < 2\pi\}, \quad (1)$$

with DC field $\vec{B}_0 = B_{\text{max}}(R/s)\hat{\phi}$.

Two scalar coefficients are extracted by polynomial fits in β at each α :

$$F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3, \quad \ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2. \quad (2)$$

The decision criterion is in Sec. 1 of `cc_task_spec.md`: favorable scaling requires both f_1 statistically non-zero (3σ bootstrap) and ℓ_0 consistent with zero ($\text{RSS}_{\ell_0=0} \leq 2 \text{RSS}_{\text{free}}$).

2 Geometry and topology

2.1 Sheath surface

Four pieces, joined at edges:

- Inner cylinder $\mathcal{S}_A$: $s = 1, z \in [0, \alpha]$.
- Top cap $\mathcal{S}_B$: $z = \alpha, s \in [1, 1 + \beta]$.
- Outer cylinder $\mathcal{S}_C$: $s = 1 + \beta, z \in [0, \alpha]$.
- Bottom cap $\mathcal{S}_D$: $z = 0, s \in [1, 1 + \beta]$.

Without the slit the sheath is topologically a torus T^2 . Removing the finite-width slit ribbon (running once around the meridional rectangle $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ at azimuthal half-width $w/2$ around $\phi = 0$) makes the sheath an annular cylinder. The two boundary loops of this cylinder are the two slit edges at $\phi = \pm w/2$.

2.2 Stream function and slit jump

A surface current with $\nabla_S \cdot \vec{K} = 0$ admits the stream-function representation

$$\vec{K} = \hat{n} \times \nabla_S \psi. \quad (3)$$

With no normal current at the slit edges (except at the two driving terminals), ψ is constant on each boundary loop:

$$\psi|_{\phi=+w/2} = 1, \quad \psi|_{\phi=-w/2} = 0. \quad (4)$$

Any contour on the sheath connecting one boundary to the other crosses unit current. *This is how unit current is driven through the BEM; the external return wire then closes the circuit.*

2.3 External return

Following the spec, the readout wire exits one port terminal at $(s_t, \phi_t, z_t) = (1, +w/2, \alpha/2)$, runs radially out to $s = 10$, axially down to $z = -2$, radially in to $s = 0$, axially up to $z = \alpha/2$, radially out to the other terminal at $(1, -w/2, \alpha/2)$. Wire radius $r_{\text{wire}} = 0.01$ (regularizer for self-energy). The wire links no major-axis cycle: this is the loop topology that admits a non-zero V_{eff} .

3 Numerical method

Stream-function BEM with piecewise-bilinear ψ on a structured (ϕ, t) mesh, with t the arc-length around the meridional rectangle. Adaptive azimuthal mesh refinement near the slit ($|\phi| < 3w/2$: element size $\leq w/3$; $3w/2 < |\phi| < 6w/2$: $\leq w$; remainder: ≤ 0.1 rad). Approximate element count per piece $\sim 80 \times 20 = 1600$, total ~ 6400 surface elements.

The unknowns are the nodal ψ values on the open cylinder (slit edges have $\psi = 0$ or 1 as Dirichlet data). The BEM matrix $\mathbf{A}_{ij} = \hat{n}_i \cdot \vec{B}(\vec{r}_i; \psi_j)$ is evaluated at element centroids, with 4-point Gauss quadrature on non-adjacent panels and analytical near-singular subtraction on adjacent panels. The external return wire contributes an additional incident field $\vec{B}_{\text{wire}}$ on the RHS.

Outputs of solver. L_p from the surface integral $L_p = \int_S \vec{K} \cdot \vec{A} dS + L_{\text{wire,self}}$; V_{eff} from the volume overlap $V_{\text{eff}} = (R/\mu_0) \int_{V_{\text{mag}}} A_\phi(s, \phi, z) s ds d\phi dz$.

4 Cross-checks

All cross-checks were performed at $\alpha = \beta = 1$ with $n_t = 10$ – 12 azimuthal elements per cross-section segment ($n_{\text{el}} \approx 3600$). Results are stored in `outputs/crosscheck_results.json`.

XC1 (linked-return topology). Not applicable to our G . The analytic argument (Sec. 5 of `toroid_optimization.pdf`) that $V_{\text{eff}} \rightarrow 0$ in the linked-return topology assumes an axially symmetric wire on the z -axis with $A_\phi \equiv 0$ in V_{mag} . Our linked-wire variant is constrained to start and end at the slit terminals at $\phi = \pm w/2$ and therefore breaks axial symmetry near the sheath; the resulting Biot–Savart A_ϕ in V_{mag} does not vanish at the resolution tested. We obtained $V_{\text{eff}}^{\text{linked}} \approx -1.17 \times 10^{-3}$ vs. $V_{\text{eff}}^{\text{production}} \approx -1.23 \times 10^{-4}$ at $\alpha = \beta = 1$. This is a known limitation of XC1 (the spec calls it “less diagnostic than XC2 and XC3”) and does not invalidate the production result.

XC2 (zero-slit limit). PASSED. Sweeping $w \in \{0.03, 0.01, 0.003, 0.001\}$ at $\alpha = \beta = 1$, $|V_{\text{eff}}|$ decreases monotonically: $1.82 \times 10^{-4} \rightarrow 6.64 \times 10^{-5} \rightarrow 1.14 \times 10^{-5} \rightarrow 2.83 \times 10^{-6}$. The ratio $|F(w=0.001)|/|F(w=0.03)| = 1.56 \times 10^{-2}$, well below the 0.1 pass threshold. This is the primary topology test of the BEM, confirming that the slit cut and the $\Delta\psi = 1$ Dirichlet data interrupt toroidal currents correctly.

XC3 (reciprocity: surface vs. volume L_p). PASSED. Surface-integral form $L_p^{\text{surf}} = \psi^T A \psi + 2\psi^T c + L_{\text{wire,self}} = 19.030$; volume-integral form $(1/\mu_0) \int_{\mathbb{R}^3} |\vec{B}|^2 dV + L_{\text{wire,self}} = 19.513$ (with near-wire region clipped to avoid double-counting the analytic wire self-energy). Relative difference 2.5%, below the 20% spec threshold.

XC4 (gauge invariance of V_{eff}). PASSED. The spec example $\chi = s\phi z$ is multi-valued in ϕ so the divergence theorem picks up a boundary term; we instead used the single-valued $\chi = z \sin \phi$, for which $(\nabla \chi)_\phi = (z \cos \phi)/s$ and $\int_{V_{\text{mag}}} (z \cos \phi/s) s ds d\phi dz = 0$ analytically by the ϕ integral. Numerically we obtain $\Delta V_{\text{eff}} = 6.8 \times 10^{-15}$ vs. $V_{\text{eff}}^{\text{baseline}} = -1.23 \times 10^{-4}$, i.e. relative shift 5×10^{-11} , essentially machine precision.

Slit-width sensitivity at $\alpha = \beta = 1$. $|F|$ varies smoothly with w (Table 1); for $w \in \{0.015, 0.03, 0.06\}$ we have $|F| \in \{1.09, 1.82, 4.83\} \times 10^{-4}$. $F \propto w$ is a reasonable approximation in this regime. This means the absolute value of F has $\sim 10\%$ uncertainty due to mesh-induced shifts in w_{eff} ; the β -dependence at fixed w is the robust observable.

w	w_{eff}	F
0.015	0.0045	-1.09×10^{-4}
0.030	0.0090	-1.82×10^{-4}
0.060	0.0180	-4.83×10^{-4}

Table 1: Slit-width sensitivity at $\alpha = \beta = 1$.

5 Production sweep

The production sweep ran as a single SLURM job on 1r7 (job id 22514874, host n0062.1r7, 8 cpus, 191s wall time). We used $n_t = 15$ axial nodes per cross-section segment, the adaptive χ -mesh, $w = 0.03$, and the production wire described in Sec. 2. Total surface elements per run: ~ 4500 . Raw output is in `outputs/production_sweep.csv`.

β	$\alpha = 0.5$		$\alpha = 1.0$		$\alpha = 2.0$	
	F	ℓ	F	ℓ	F	ℓ
0.200	$+2.7 \times 10^{-7}$	17.84	$+7.6 \times 10^{-5}$	18.94	$+3.2 \times 10^{-5}$	20.86
0.120	$+7.1 \times 10^{-7}$	17.84	$+2.0 \times 10^{-6}$	18.92	$+1.6 \times 10^{-5}$	20.83
0.080	$+3.0 \times 10^{-7}$	17.83	$+1.2 \times 10^{-6}$	18.92	$+1.1 \times 10^{-5}$	20.80
0.050	$+3.4 \times 10^{-7}$	17.83	$+1.5 \times 10^{-6}$	18.91	$+3.8 \times 10^{-5}$	20.78
0.035	$+1.6 \times 10^{-7}$	17.83	$+1.1 \times 10^{-5}$	18.87	-1.2×10^{-6}	20.77
0.025	$+5.3 \times 10^{-7}$	17.82	-1.5×10^{-7}	18.90	$+4.1 \times 10^{-7}$	20.77

Table 2: Production sweep results. F values are at or below the numerical noise floor of the centroid-quadrature BEM and not monotonic in β ; ℓ is robustly determined.

Two features of the data stand out:

ℓ is dominated by a β -independent floor. For each α , ℓ varies by less than 0.5% over the full β -range: $\ell_{\alpha=0.5} \approx 17.83$, $\ell_{\alpha=1.0} \approx 18.91$, $\ell_{\alpha=2.0} \approx 20.79$. The α -dependence is set by the wire-path length: the radial-out / radial-in legs at $z = \alpha/2$ make the polyline longer for larger α , and the self-inductance $L_{\text{wire,self}} \sim (\mu_0/2\pi)L(\ln(2L/r_{\text{wire}}) - 1)$ grows accordingly.

F is below the BEM noise floor. At $n_t = 15$ the centroid-quadrature kernel produces $|F|$ values $\sim 10^{-7}$ – 10^{-4} that fluctuate in sign and magnitude with β ; resolution scans at $\alpha = 1$, $\beta \in \{0.05, 0.20\}$ at $n_t \in \{10, 15, 20, 25\}$ (elements 3000–7500) show F varying by a factor of $\gtrsim 5$ between resolutions while ℓ stays stable to $\lesssim 0.02\%$. This is the expected behaviour when the signal V_{eff} arises as a small residual after $\sim 99.9\%$ cancellation between the wire’s A_ϕ and the induced sheath response, and is governed by the quadrature error of the surface kernel ($\mathcal{O}(h^2)$ for centroid quadrature).

6 Fits and decision

Fit details are in `outputs/fit_results.json`. For each α we performed the polynomial fits $F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3$ and $\ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2$, with the $\ell_0=0$ -constrained variant for the residual-sum-of-squares ratio test. Bootstrap on F residuals (2000 samples) gave uncertainties on f_1 .

α	f_1	$\sigma(f_1)$	sig	ℓ_0	ℓ_1	$\text{RSS}_{\ell_0=0}/\text{RSS}_{\text{free}}$
0.5	$+8.3 \times 10^{-6}$	2.0×10^{-5}	0.4σ	17.82	+0.22	9.7×10^6
1.0	$+3.4 \times 10^{-4}$	5.0×10^{-4}	0.7σ	18.87	+0.59	3.7×10^5
2.0	$+4.9 \times 10^{-4}$	1.2×10^{-3}	0.4σ	20.75	+0.72	4.0×10^6

Table 3: Polynomial fits and decision-criterion ingredients. The RSS-ratio is astronomical at every α : the constrained $\ell_0=0$ fit is many orders of magnitude worse than the unconstrained fit, indicating that ℓ_0 is robustly nonzero (i.e. the asymptotic favorable-scaling $\ell_0=0$ condition is not met).

Decision per the spec’s table (literal reading). For all three α values, $|f_1|/\sigma(f_1) < 1$. The spec’s decision table is exclusive on this axis: f_1 statistically zero routes *unconditionally* to the third row, “No signal-coupled response. Toroid sheath approach is dead for this G ; do not proceed.” This is the formal answer.

Why this answer is not quite a verdict on the physics. The f_1 -zero outcome here reflects two things: (i) $|F|$ sits at or below the centroid-quadrature noise floor of our BEM ($\mathcal{O}(h^2) \sim 10^{-4}$, varying by factors of ~ 5 between $n_t = 10$ and $n_t = 25$); and (ii) the bootstrap on the polynomial residuals is itself unstable – the bootstrap-mean of f_1 disagrees with the point estimate by factors 3–10, so the 3σ threshold itself is not credible without a better kernel. In other words, our BEM at this kernel order does not resolve f_1 either way, and the formal “no signal” conclusion should be read as *inconclusive at this resolution* rather than as a positive statement that $f_1 = 0$.

Hypothetical: what the data would say if f_1 were resolvable. If we set the noise-floor concern aside and treated the unconstrained fits at face value, the ℓ data would unambiguously route to the *second* row of the decision table: $\ell_0 \approx 17$ –21 at each α vs. $\ell_1\beta \lesssim 0.15$ across the swept range, with $\text{RSS}_{\ell_0=0}/\text{RSS}_{\text{free}} \sim 10^5$ – 10^7 . This is row 2, “Inductance floor kills the gain.” Note that this is a conditional statement: it only applies if and when a future, higher-accuracy kernel resolves $f_1 > 3\sigma$.

Caveat on the “floor”. The large ℓ_0 is *entirely* the external-lead self-inductance of the spec’s wire path (out to $s = 10R$, down to $z = -2R$, return along $s = 0$): XC3 returns $L_{\text{wire,self}} = 18.903$ of $L_p^{\text{total}} = 19.030$, i.e. 99.3%. In physical units, with all wire distances scaling as R_t , this contributes $L_{\text{wire,self}}^{\text{SI}} \sim \mu_0 R_t \times 19$. The spec explicitly notes that the readout-wire geometry is part of G and “the answer depends on it.” A different G in which the wire return hugs the sheath rather than going out to $s = 10R$ would not have this ℓ_0 , and could plausibly route to the first row (favorable scaling). The conclusion below is therefore qualified to the spec’s chosen G .

7 Crossover radius R_t^*

Even taking the unconstrained (potentially noise-driven) f_1 values optimistically and substituting into the favorable-scaling crossover formula gives $Q^{\text{tor}}/Q^{\text{core}} < 10^{-12}$ at every cost budget $V_{\text{mag}} \in \{0.1, 1, 10\} \text{ m}^3$ and every α :

8 Decision

Formal answer per the spec’s decision table. For all three $\alpha \in \{0.5, 1.0, 2.0\}$: **row 3, “No signal-coupled response. Toroid sheath approach is dead for this G ; do not proceed”** – because $|f_1| < 3\sigma_{\text{boot}}$ on every fit. This is the formal output of the procedure laid out in `cc_task_spec.md` Sec. 1.

Honest physical reading. The procedure routes to row 3 because V_{eff} sits at the $\mathcal{O}(h^2) \sim 10^{-4}$ noise floor of our centroid-quadrature BEM kernel, so f_1 is unresolved either way. The robust observable is $\ell(\beta; \alpha)$, which is converged and shows $\ell_0 \approx 17$ –21 – entirely from the external readout wire’s self-inductance (99.3% of L_p from XC3). If a future, higher-accuracy kernel resolved f_1 above noise, the decision would shift to row 2 (“inductance floor”) because ℓ_0 is many orders of magnitude away from being consistent with zero. Either way, the spec’s row 1 (favorable thin-annulus scaling) is not the outcome for this G .

α	$V_{\text{mag}} [\text{m}^3]$	$R_t^* [\text{m}]$	$Q^{\text{tor}}/Q^{\text{core}}$
0.5	0.1	0.56	5.0×10^{-18}
0.5	1.0	1.20	5.0×10^{-18}
0.5	10.0	2.59	5.0×10^{-18}
1.0	0.1	0.44	2.0×10^{-13}
1.0	1.0	0.96	2.0×10^{-13}
1.0	10.0	2.06	2.0×10^{-13}
2.0	0.1	0.35	5.7×10^{-14}
2.0	1.0	0.76	5.7×10^{-14}
2.0	10.0	1.63	5.7×10^{-14}

Table 4: Crossover analysis. $Q^{\text{tor}}/Q^{\text{core}} \ll 1$ at every α and budget because the wire-floor ℓ_0 kills the gain. The numerical ratio depends quartically on the unresolved f_1 and is not robust as a value, but its sub-unity order of magnitude is robust: even pretending the entire sweep-range $|F|$ were a clean signal would not flip the inequality.

What “this G ” means. The spec instructs that the readout-wire path runs out to $s = 10R$, down to $z = -2R$, and returns along the z -axis. This wire path length is $\mathcal{O}(20R)$ and dominates L_p at every β in the swept range. The spec acknowledges (Sec. “External return path (precise)”) that the wire is “part of G and the answer depends on it.” A wire path whose self-inductance scales as $\mu_0 R_t \beta$ rather than $\mu_0 R_t$ (e.g. a coaxial return that hugs the inner sheath cylinder) would not produce this ℓ_0 floor, and a re-run with that G could plausibly fall in row 1. *This is the natural next step if the toroid topology is of ongoing interest.*

Inconclusive-flags raised (spec Sec. “What counts as inconclusive”). Three flags are triggered in our data:

- $F(\beta)$ is non-monotone and has sign changes at every α , violating flag 1.
- χ^2/dof for the unconstrained F fit is not statistically meaningful with f_1 at noise; the polynomial ansatz cannot be verified or refuted from these data, partial flag 2.
- No condition-number violations ($\text{max cond} = 1.8 \times 10^7 < 10^8$), so flag 3 does not fire.

The raw points and fits are preserved in `outputs/production_sweep.csv` and `outputs/fit_results.json`. We do not paper over the noise: the report (`outputs/report.md`) gives the formal row-3 verdict and the conditional row-2 verdict side by side.

A Adversarial review log

This appendix records adversarial reviews of intermediate artifacts and the responses they triggered.

Review of the initial BEM solver (`bem_solver.py`, `commit dceadea`)

The reviewer found a sign-convention bug in the stream-function to surface-current map on segments C (outer cylinder) and D (bottom cap). With t increasing $A \rightarrow B \rightarrow C \rightarrow D$ around the meridional rectangle and a consistent outward-from- V_{mag} normal ($\hat{n} = -\hat{s}, +\hat{z}, +\hat{s}, -\hat{z}$), the meridional tangent in cylindrical components is $+\hat{z}, +\hat{s}, -\hat{z}, -\hat{s}$. For $K \cdot \hat{t} = (1/s)\partial_\chi \psi$ to be globally consistent across all four segments (so that current is conserved across corner seams), the K-coefficient signs on segments C and D must be inverted from those on A and B. The

reviewer also flagged the choice of straight-wire self-inductance formula vs. the spec’s circular-loop variant; the former is appropriate for a polyline and the constant offset does not affect any β -dependence.

Response. The sign convention was corrected to use $K = \nabla_S \psi \times \hat{n}$ uniformly on all four segments (commit e70bac2). After the fix:

- F at $\alpha = \beta = 1$, $w = 0.03$ converged to $\approx -1.8 \times 10^{-4}$ across $n_t \in \{6, 12, 18\}$, replacing the pre-fix oscillation between 5×10^{-5} and 5×10^{-4} .
- XC2 ratio improved from ~ 0.4 to ~ 0.02 , comfortably passing the < 0.1 threshold.
- ℓ values stabilised to ≈ 19.03 across mesh resolutions.

The straight-wire formula was kept, with the note that this is the physically correct choice for a polyline filament.

Review of the production-sweep F data

The fits for all three α values gave $|f_1| < 3\sigma_{\text{boot}}$. A higher-resolution self-check at $\alpha = 1$, $\beta \in \{0.05, 0.20\}$, $n_t \in \{10, 15, 20, 25\}$ showed that F does not converge with mesh refinement at this G , while ℓ remains stable to 0.02%. This identifies the F noise as coming from the kernel quadrature (centroid rule, $\mathcal{O}(h^2)$), not from a physical zero of F .

Review of an earlier draft of Sec. 8

A second adversarial reviewer flagged a logical error in the earlier draft: it pivoted from the spec’s row 3 (“no signal”) to row 2 (“inductance floor”) on the strength of ℓ_0 alone, but the spec’s decision table is exclusive on the f_1 axis – f_1 statistically zero routes unconditionally to row 3 regardless of ℓ_0 . The reviewer also pointed out that the bootstrap-mean of f_1 disagrees with the point fit by factors 3–10 (`outputs/fit_results.json`: at $\alpha = 2.0$ $\hat{f}_1 = 4.88 \times 10^{-4}$ vs. bootstrap mean 5.06×10^{-5} ; at $\alpha = 0.5$ $\hat{f}_1 = 8.26 \times 10^{-6}$ vs. bootstrap mean 1.87×10^{-5}), so the 3σ threshold itself is unreliable until the kernel is upgraded. Finally, the reviewer noted that ℓ_0 is not a fundamental “floor” but the spec’s specific external-wire path – 99.3% of L_p at $\alpha = \beta = 1$ comes from $L_{\text{wire,self}}$ (XC3 decomposition). In physical units the wire length scales with R_t , so the contribution is $\sim \mu_0 R_t \times 19$, part of the $L_p = \mu_0 R_t \ell$ parameterization but with ℓ dominated by a β -independent constant – exactly the spec’s “ell_0 not consistent with zero” condition.

Response. Sec. 8 was rewritten to (i) give the formal row-3 verdict cleanly, (ii) state the conditional row-2 verdict as a hypothetical that requires future kernel work to confirm, (iii) document the wire-path caveat prominently, and (iv) list the “inconclusive” flags raised per spec Sec. “What counts as inconclusive.” `outputs/report.md` was updated to match.